UTTARAKHAND

- The state has **13 glacial lakes officially identified as sensitive**, including concentrations in Pithoragarh and Chamoli
- The vulnerable river systems include the **Alaknanda/ Dhauliganga, Bhagirathi, Mandakini and Kali/ Sharda**
- What makes Uttarakhand particularly dangerous is that pilgrimage towns, roads, bridges, hotels, hydro projects and villages have expanded along very confined valleys

SIKKIM

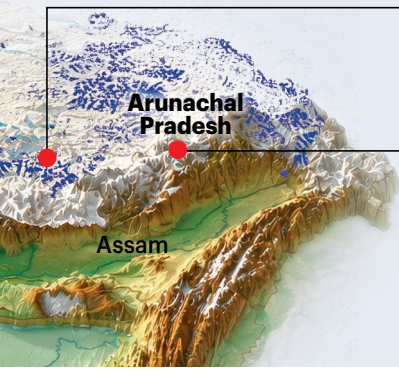
- An estimated **17 'very high risk' glacial lakes** are in Sikkim. Another Glacial Lake Outburst Flood (GLOF) pouring into the Teesta rivers system like the one in South Lhonak Lake did in 2023 can become a cascading infrastructure disaster of enormous magnitude

TRANS-HIMALAYAN DANGER FROM TIBET, NEPAL AND BHUTAN

- Many headwaters of the **Sutlej, Brahmaputra and Himalayan tributaries begin in Tibet; others enter India through Nepal or Bhutan.** A lake or glacier collapse across the border can send a flood into India long before Indian authorities have enough information to assess it

ARUNACHAL PRADESH

- The danger comes both from **lakes inside Arunachal** and from the huge Tibetan catchments upstream of the **Siang/ Brahmaputra, Subansiri and other rivers**
- IIT Guwahati researchers have identified **492 locations in the eastern Himalaya where new glacial lakes could potentially form** as glaciers retreat



tributaries. These are steep gorges, with hydropower stations, reservoirs, highways and settlements directly below glaciers and unstable slopes. State surveys indicate a rapid increase in glacial and high-altitude water bodies, while Ghepang Ghat in Lahaul-Spiti shows how dramatically a single lake can expand. The Sutlej also enters India from Tibet, making upstream information indispensable.

BORDER BLINDSPOTS

Ladakh and Jammu & Kashmir have fewer densely populated valleys, but that does not make them safe. Communities, roads, military facilities and tourism infrastructure are concentrated along the Indus, Zanskar, Shyok, Nubra and Chenab systems that flow in and out of China and Pakistan-occupied Kashmir. Monitoring is difficult, communications sparse and evacuation alternatives limited. Thawing permafrost can release a rock-ice avalanche or create a temporary river dam; its failure can send a surge through a settlement that had no recognised glacial lake above it. At the other extreme border of India, Arunachal Pradesh is the least understood of India's major danger zones and potentially among the most consequential. Rapidly expanding lakes have been identified in the Dibang Valley and the Mago Chu basin of Tawang. Large hydropower ambitions—33 major projects to generate 19,165 MW slated to come up in the next 10 years—magnify downstream exposure. The state is also vulnerable to events in the vast Tibetan catchments of the Siang-Brahmaputra and its tributaries, where India has little real-time visibility.

The risk does not end at the Himalayan states. Northern West Bengal lies below Sikkim on the Teesta. Assam and the wider Brahmaputra valley sit downstream of Tibetan glacial terrain. Bihar is repeatedly exposed through Himalayan rivers such as the Kosi and Gandak, where glacial hazards can combine with monsoon rainfall, landslides and embankment failures. A high-mountain event, therefore, can propagate into the plains, carrying sediment, damaging barrages and bridges and displacing people far from its source.



"Our exchange mechanisms with our neighbours are at a very small level. We treat borders as geometric lines instead of shared ecological spaces"

MAHENDRA LAMA, *Senior Advisor, ICIMOD*

The Nepal catastrophe knocked multiple generating and transmission facilities offline. Yet environmental impact assessments and dam design have conventionally relied on the past behaviour of rivers—historical rainfall, recorded floods and familiar lake scenarios—in a mountain system whose future is moving outside that record. "We are creating infrastructure based on the past knowledge of these valleys, whereas we need to create it based on the future of these valleys," says Haritashya. Not every Himalayan valley must be declared off limits, he argues, but not every valley is suitable for a large dam, highway or town either. The test should be whether the project remains safe over its entire lifetime under projected warming, glacier retreat, permafrost loss and extreme precipitation and what its failure would do to every community below it.